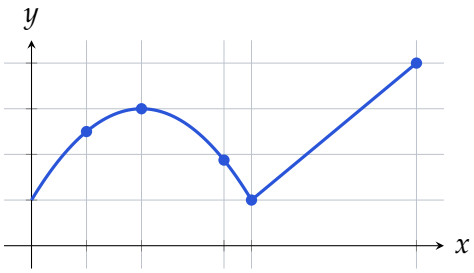
Lesson 1-8 • MVT

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: recognize when the Mean Value Theorem applies, and use it.

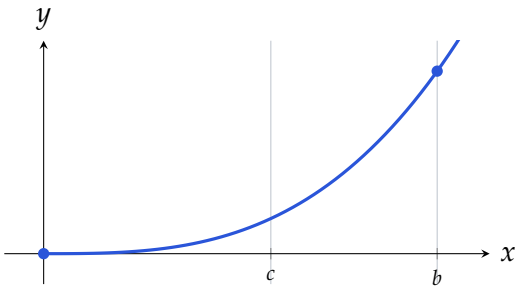
Review: differentiability

What is the instantaneous rate of change at each marked point: positive, negative, 0 or DNE?



$x = 1$: _____
$x = 2$: ____
$x = 3.5$: _____
$x = 4$: _____
$x = 7$: _____

The Mean Value Theorem (MVT)

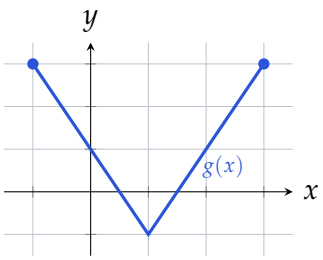


Needs	If f is _____ on $[a, b]$ and _____ on (a, b),
Proves	then there is a c in (a, b) with _____.

In words: the slope of the _____ at some point equals the slope of the _____ (the average rate of change). The MVT is about _____.

Example 1

f is a differentiable function with $f(2) = 5$ and $f(7) = 20$. Explain why there must be a value c, $2 < c < 7$, with $f'(c) = 3$.

Example 2	 $g(x) = 2\|x-1\| - 1$ on $[-1,3]$. Find the average rate of change of g on $[-1,3]$. Does the MVT guarantee a c in $(-1,3)$ with $g'(c)$ equal to it?
Summary	